

Talos

Spoken instructions for
learned dinner-table tasks

Two SO-101 arms
Six neural dinner skills
Bottle relays in both directions



Intel online challenge + Speechmatics integration

Six learned skills, one sequence

“Set the table”

Place the bottle, plate and mug.
Open the drawer.
Retrieve and place fork and spoon.

8/10 frozen full-sequence trials.
Task-preset variation; fixed destinations.



Speech and control

Supported voice commands use Speechmatics

A human "Set the table" command completed all six skills.
The local voice run included 15 live mug corrections.

Bounded grammar and RGB scene checks choose the next skill.
Unsupported commands are refused. Progress and cancellation stay visible.

The hosted demo accepts typed commands. The local browser lab supports voice.

Learned bottle relay in both directions

Each arm releases onto the shared table

- 1 First arm lifts and releases the bottle on the table.
- 2 Both arms park before the second grasp.
- 3 Second arm places the bottle at its trained destination.

Frozen trials: 5/5 left to right and 5/5 right to left.

Two upright start regions and fixed goals. No airborne transfer.

How learned control works

Neural skills with bounded visual correction

The five newer dinner skills use 201 verified inputs.
Four bottle-relay models use 52 inputs; originals are preserved.

Joint feedback regulates neural motion.
A separate physical monitor checks contact, placement and parked arms.

Late mug vision: 20/20 live and 20/20 baseline workflows.
With corrective images frozen: 0/20. Narrow tested starts.

Physical evaluation

Frozen physical trials with every outcome retained

8 / 10

Six learned dinner tasks
Randomized task-preset starts

10 / 10

Bottle relay trials
Five in each direction

Dinner failures: mug 14.59 mm and spoon 13.61 mm XY error.
Fixed goals and small task-preset variations define this result.

Intel execution and remaining eligibility

0.157 ms

Core i7-10850H CPU
FP32 neural inference

0.688 ms

Intel UHD integrated GPU
FP32 neural inference

Six-skill baseline runs with Intel CPU/iGPU and UHD rendering.
Core Ultra Series 2/3 and the latest live-mug Intel run remain open.

0.157/0.688 ms describe the original bottle network only.
Legacy hardware execution does not establish Core Ultra eligibility.

Reproducible baseline, active development

Code, models, compact inputs and complete outcomes

Private CPU demo: all six skills passed in 287.79 wall seconds.
All frozen outcomes and compact training inputs remain packaged.
Original work is MIT; third-party components retain their licenses.

Seven jointly randomized items: no complete workflow has passed.
The adapted CLIPort runtime works; its Talos action policy is untrained.

github.com/jannissio/talos-ai-infra